

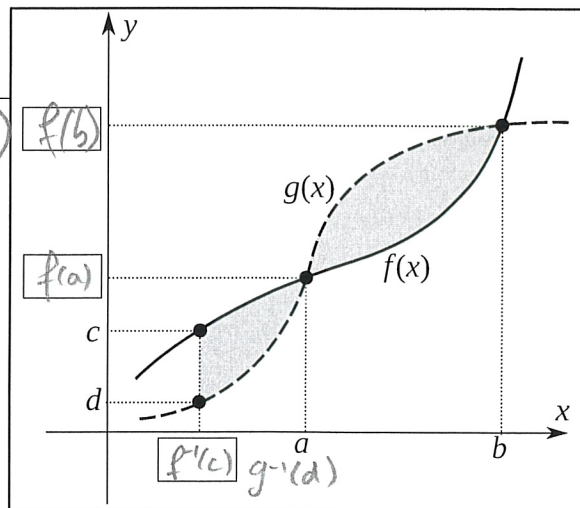
Do NOT write ANYTHING above this line!

1. (6 pts) Write the integrals for the area defined by the shaded region.

(a) On the plot: fill the empty boxes.

(b) A_x : Write area as integral(s) with respect to x and

(c) A_y : Write area as integral(s) with respect to y .



① $[-1/2]$
for each wrong
for a max
of -1

$$A_x = \int_{f^{-1}(c)}^a (f(x) - g(x)) dx + \int_a^b (g(x) - f(x)) dx$$

$$A_y = \int_d^c (g^{-1}(y) - f^{-1}(c)) dy + \int_c^{f(a)} (g^{-1}(y) - f^{-1}(y)) dy + \int_{f(a)}^{f(b)} (f^{-1}(y) - g^{-1}(y)) dy$$

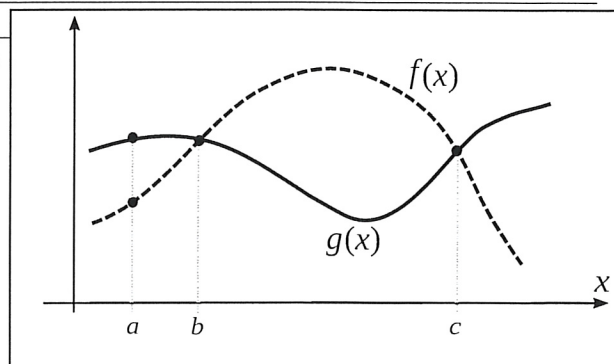
2. (4 pts) Applications of integrals: averages.

For the functions depicted to the right, write the corresponding integrals for the average vertical distance between the graphs of the functions. Perform this for the following intervals:

(a) Average vertical distance on $a \leq x \leq b$

(b) Average vertical distance on $b \leq x \leq c$

(c) Average vertical distance on $a \leq x \leq c$



① or one could use $| \cdot | \rightarrow \frac{1}{c-a} \int_a^c |f(x) - g(x)| dx$

(a) ave. on $[a, b]$: $\frac{1}{b-a} \int_a^b (g(x) - f(x)) dx$

(b) ave. on $[b, c]$: $\frac{1}{c-b} \int_b^c (f(x) - g(x)) dx$

② (b) ave. on $[a, c]$: $\frac{1}{c-a} \left(\int_a^b (g(x) - f(x)) dx + \int_b^c (f(x) - g(x)) dx \right)$

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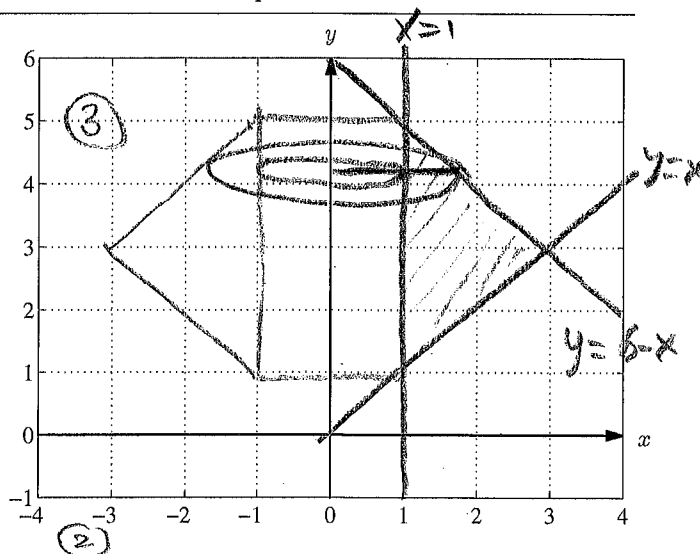
3. (10 pts) Using **WASHERS** write an integral (or integrals) for volume of the solid generated by rotating about the **y-axis** the region inside the graphs of the following functions: $x = 1$, $y = f(x) = x$, and $y = g(x) = -x + 6$. Sketch (i) the solid, (ii) the region, and (iii) a typical washer for this object.

NOTE: you only need to write the integral(s) but you do not need to compute it!

$1 \leq y \leq 5$ $r_o: y = x \Rightarrow x = f^{-1}(y) = y$
 $r_i: x = 1$

$3 \leq y \leq 5$ $r_o: y = 6 - x \Rightarrow x = f^{-1}(y) = 6 - y$
 $r_i: x = 1$

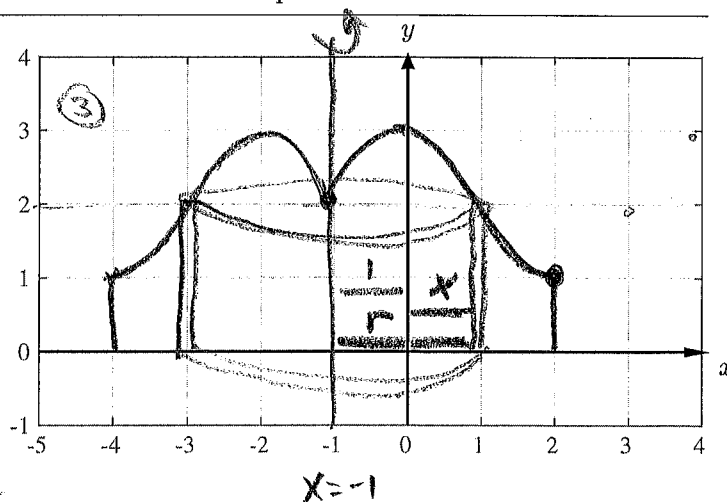
$$V = \int_{y_1}^{y_2} \pi (r_o^2 - r_i^2) dy \quad (1)$$



$$V_1 = \int_1^3 \pi [y^2 - 1] dy + \int_3^5 \pi [(6-y)^2 - 1] dy$$

4. (10 pts) Using volumes by **SHELLS**, write an explicit integral for the solid generated by rotating about the line $x = -1$ (note that the line is off-axis!) the region delimited by the graphs of: $y = 2 + \cos(\frac{\pi}{2}x)$ and $y = 0$ for $-1 \leq x \leq 2$. (a) Sketch the functions, (b) the region, (c) the solid and (d) a typical shell for this object.

Note: you only need to write the integral but you do not need to compute it!



$$V = \int_{x_1}^{x_2} \text{arc. height} \cdot dx \quad (1)$$

$$= \int_{-1}^2 2\pi r \cdot h(x) dx$$

$$1+x=r \quad (1)$$

$$= \int_{-1}^2 2\pi (1+x) [2 + \cos(\frac{\pi}{2}x)] dx$$

$$V_2 = \int_{-1}^2 2\pi (1+x) [2 + \cos(\frac{\pi}{2}x)] dX$$

Do NOT write ANYTHING above this line!

5. (8 pts) (a) Filling the boxes below, describe the volume of revolution that leads, using the method of **WASHERS**, to the integral:

$$I = \int_0^4 \pi [(x+2)^2 - 4] dx.$$

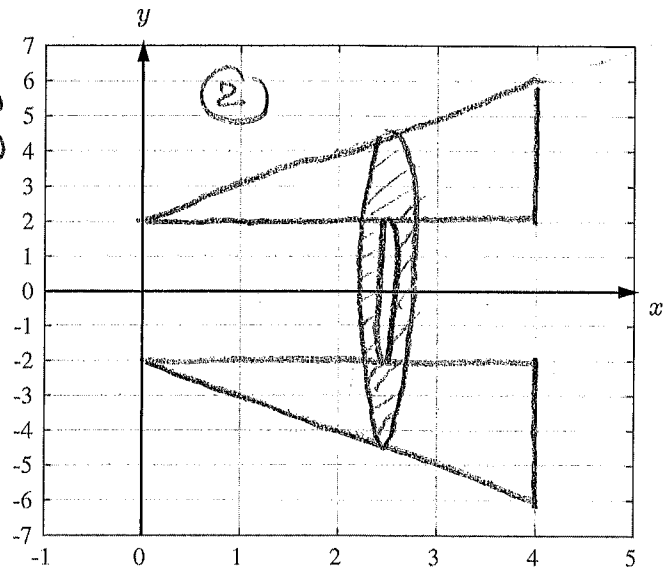
$$\begin{aligned} r_o &= x+2 & (1) \\ r_i &= 2 & (1) \end{aligned}$$

- (b) Sketch the functions, the solid and
(c) a typical washer for this object.

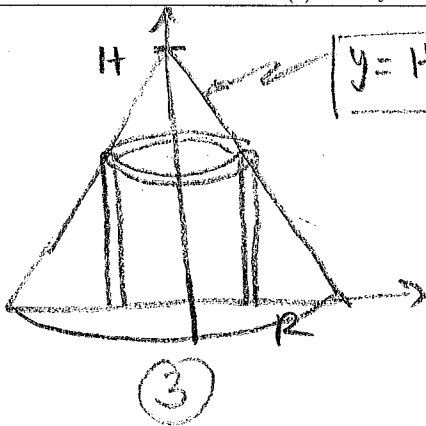
The integral I above describes the volume generated by rotating about the $y = 0$ (1) axis, the region

delimited by the functions $y = x+2$ (1) and

$$y = 2 \quad (1) \quad \text{for} \quad 0 \leq x \leq 4 \quad (1)$$



6. (10 pts) Show, using the method of **VOLUME BY SHELLS**, that the volume of a cone with circular base of radius R and height H is given by $V = \frac{1}{3}\pi R^2 H$. Draw a diagram including a typical slice for this object. Clearly indicate the function(s) that you are plotting and the interval of integration.



$$y = H - \frac{H}{R}x \quad (1)$$

$$V = \int_0^R \text{circ. height}$$

$$= \int_0^R 2\pi x \cdot h \, dx \quad (1)$$

$$= \int_0^R 2\pi x \cdot \left(H - \frac{H}{R}x\right) dx$$

$$\text{OR} \quad = 2\pi H \int_0^R x \left(1 - \frac{x}{R}\right) dx$$

$$= 2\pi H \int_0^R \left(x - \frac{x^2}{R}\right) dx = 2\pi H \left[\frac{x^2}{2} - \frac{x^3}{3R}\right]_0^R$$

$$= 2\pi H \left[\frac{R^2}{2} - \frac{R^3}{3R}\right] = 2\pi H R^2 \left[\frac{1}{2} - \frac{1}{3}\right]$$

$$= 2\pi H R^2 \left[\frac{3}{6} - \frac{2}{6}\right] = 2\pi H R^2 \left[\frac{1}{6}\right]$$

$$\boxed{V = \frac{1}{3}\pi R^2 H}$$

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7. (10 pts) Work.

- a) (3 pts) Compute the work done by the force $F(x) = kx$ (i.e., a linear, Hooke's, spring) when moving an object from $x = x_0$ to $x = x_1$.

$$W = \int_{x_0}^{x_1} F(x) dx = \int_{x_0}^{x_1} kx dx = \left[\frac{kx^2}{2} \right]_{x_0}^{x_1} = \frac{k}{2}(x_1^2 - x_0^2)$$

(1) (2)

- b) (3 pts) If one can only afford to use 12 Joules (Joule is the metric unit for energy: $1 \text{ J} = 1 \text{ kg m}^2/\text{s}^2$), how far is it possible to drag an object, starting at $x = 0$ whose drag force is given by $F(x) = 4 + 2x$? Simplify as much as possible and leave your result as a single number with the correct units.

$$W = 12 \Rightarrow \int_0^D F(x) dx = 12 \Rightarrow \int_0^D (4 + 2x) dx = 12 \Rightarrow [4x + x^2]_0^D = 12$$

$$\Rightarrow 4D + D^2 - 12 = 0 \Rightarrow D = \frac{-4 \pm \sqrt{16 + 48}}{2} = \frac{-4 \pm \sqrt{64}}{2}$$

$$\Rightarrow D = \frac{-4 \pm 8}{2} = \frac{1}{2}(-4 - 8, -4 + 8) = \frac{1}{2}(-12, 4) = (-6, 2)$$

Since move to right $\Rightarrow D = 2 \text{ m}$ (1)

- c) (4 pts) A sack with 30 Kg of sand is being lifted from $y = 0 \text{ m}$ to $y = 20 \text{ m}$. The sack has a small hole that allows 100 g of sand to be lost every meter that the sack is pulled up. Compute the total energy necessary to lift the sack.
Hint#1: $F = m \times g$, use $g \approx 10 \text{ m/s}^2$, $1 \text{ Kg} = 1,000 \text{ g}$.
Hint#2: First compute the mass as a function of height.

$$m(y) = 30 - 0.1y \quad (1)$$

$$\begin{aligned} W &= \int_0^{20} F(y) dy = \int_0^{20} m(y) \cdot g dy \\ &= 10 \int_0^{20} (30 - 0.1y) dy = 10 \left[30y - \frac{0.1y^2}{2} \right]_0^{20} \\ &= 10 \left[600 - \frac{0.1 \cdot 400}{2} \right] = 10 \left[600 - 20 \right] \\ &= 10 \times 580 = 5800 \text{ J} \end{aligned}$$

(2) (1)

Do NOT write ANYTHING above this line!

8. (20 pts) Compute the following integrals

a) (4 pts) $I_1 = \int \cos^5(x) \sin^2(x) dx = \int \cos^4(x) \sin^2(x) dx = \int (\cos^2)^2 \sin^2(x) \frac{dx}{du}$

$$\begin{cases} du = \cos x \\ u = \sin x \end{cases} \quad (1)$$

$$\begin{aligned} &= \int (1-u^2)^2 u^2 du = \int (1-2u^2+u^4)u^2 du \\ &= \int (u^2-2u^4+u^6) du = \left[\frac{1}{3}u^3 - \frac{2}{5}u^5 + \frac{1}{7}u^7 + C \right] \quad (1) \end{aligned}$$

$$I_1 = \frac{1}{3} \sin^3 x - \frac{2}{5} \sin^5 x + \frac{1}{7} \sin^7 x + C$$

b) (5 pts) $I_2 = \int 3x^2 \sin(x) dx = -3x^2 \cos x - \int (-\cos x) 6x dx = \left[-3x^2 \cos x + 6 \int x \cos x dx \right] \quad (1)$

Parts $\begin{cases} u = 3x^2 \rightarrow du = 6x dx \\ v = -\cos x \leftarrow dv = \sin x dx \end{cases} \quad (1)$

$$\begin{cases} u = x \rightarrow du = dx \\ v = \sin x \leftarrow dv = \cos x dx \end{cases} \quad (1)$$

$$\begin{aligned} \Rightarrow I_2 &= -3x^2 \cos x + 6 \left[x \sin x - \int \sin x dx \right] \\ &= -3x^2 \cos x + 6x \sin x + 6 \cos x + C \end{aligned}$$

$$I_2 = -3x^2 \cos x + 6x \sin x + 6 \cos x + C$$

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c) (3 pts) $I_3 = \int 3x \cosh(2x^2 + 1) dx$

U-sub $\boxed{u = 2x^2 + 1} \rightarrow du = 4x dx \Rightarrow 3x dx = \frac{3}{4} du$

~~Part 5~~ $\Rightarrow I_3 = \int \cosh u \cdot \frac{3}{4} du = \boxed{\frac{3}{4} \int \cosh u du} \quad (1)$
 $= \frac{3}{4} \sinh u + C$
 $= \frac{3}{4} \sinh(2x^2 + 1)$
 (1)

Not necessary

$I_3 = \frac{3}{4} \sinh(2x^2 + 1) + C$

d) (5 pts) $I_4 = \int_0^4 x e^{x+2} dx = \left[x e^{x+2} \right]_0^4 - \int_0^4 e^{x+2} dx = \left[x e^{x+2} \right]_0^4 - \left[e^{x+2} \right]_0^4$

Parts $\boxed{\begin{matrix} u = x \rightarrow du = dx \\ v = e^{x+2} \leftarrow dv = e^{x+2} dx \end{matrix}} \quad (1)$

$\Rightarrow I_4 = \left[x e^{x+2} - e^{x+2} \right]_0^4 = \left[e^{x+2} (x-1) \right]_0^4$
 $= e^6 (4-1) - e^2 (0-1) = 3e^6 + e^2$

(2)

$I_4 = 3e^6 + e^2$

e) (3 pts) $I_5 = \int \cos^4(t) dt$

double angle : $\cos^4(t) = (\cos^2(t))^2 = \left(\frac{1+\cos 2t}{2} \right)^2 = \frac{1}{4} [1 + 2\cos 2t + \cos^2 2t]$

$\Rightarrow \boxed{I_5 = \frac{1}{4} \int (1 + 2\cos 2t + \cos^2 2t) dt} \quad (1)$

but $\cos^2 2t = \frac{1 + \cos 4t}{2}$

$\Rightarrow I_5 = \frac{1}{4} [t + \sin 2t] + \frac{1}{4} \int \frac{1}{2} (1 + \cos 4t) dt \quad (1)$
 $= \frac{t}{4} + \frac{1}{4} \sin 2t + \frac{1}{8} t + \frac{1}{4 \times 2} \sin 4t$

$\frac{1}{4} + \frac{1}{8} = \frac{2}{8} + \frac{1}{8} = \frac{3}{8}$

$I_5 = \frac{3}{8} t + \frac{1}{4} \sin 2t + \frac{1}{32} \sin 4t + C$

7

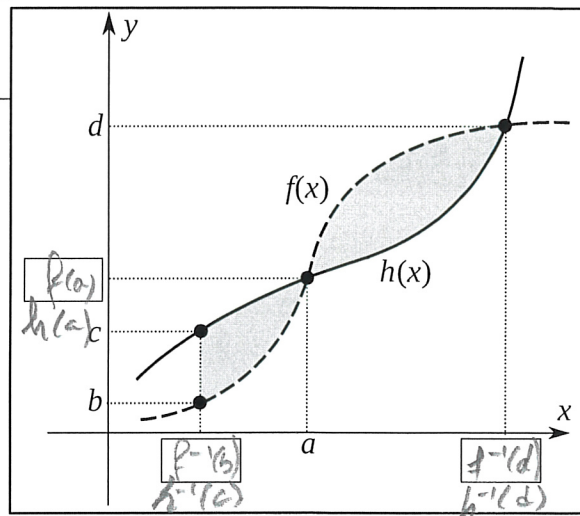
(1)

Not necessary

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1. (6 pts) Write the integrals for the area defined by the shaded region.

- (a) On the plot: fill the empty boxes.
 (b) A_x : Write area as integral(s) with respect to x and
 (c) A_y : Write area as integral(s) with respect to y .



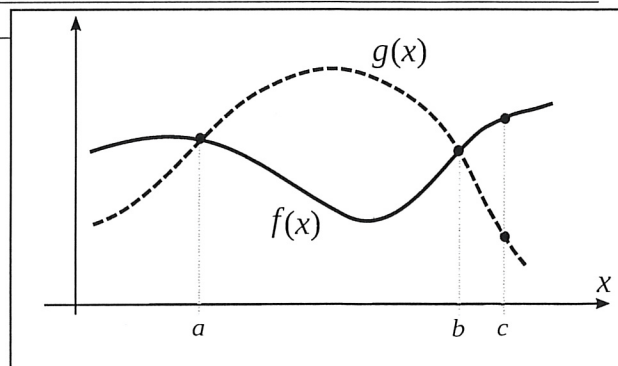
$$A_x = \int_{f^{-1}(b)}^a (h(x) - f(x)) dx + \int_a^{f^{-1}(d)} (f(x) - h(x)) dx$$

$$A_y = \int_b^c (h^{-1}(y) - f^{-1}(y)) dy + \int_c^{f(a)} (h^{-1}(y) - f^{-1}(y)) dy + \int_{f(a)}^d (h^{-1}(y) - f^{-1}(y)) dy$$

2. (4 pts) Applications of integrals: averages.

For the functions depicted to the right, write the corresponding integrals for the **average vertical distance** between the graphs of the functions. Perform this for the following intervals:

- (a) Average vertical distance on $a \leq x \leq b$
 (b) Average vertical distance on $b \leq x \leq c$
 (c) Average vertical distance on $a \leq x \leq c$



(a) ave. on $[a, b]$: $\frac{1}{b-a} \int_a^b (g(x) - f(x)) dx$

(b) ave. on $[b, c]$: $\frac{1}{c-b} \int_b^c (f(x) - g(x)) dx$

(c) ave. on $[a, c]$: $\frac{1}{c-a} \left[\int_a^b (g(x) - f(x)) dx + \int_b^c (f(x) - g(x)) dx \right]$

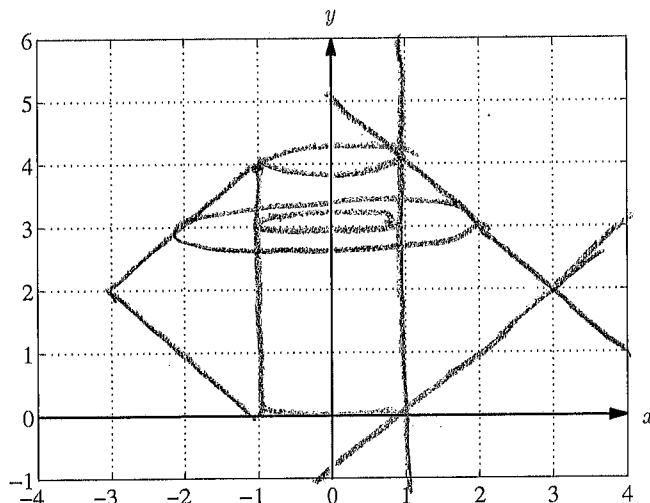
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3. (10 pts) Using **WASHERS** write an integral (or integrals) for volume of the solid generated by rotating about the y -axis the region inside the graphs of the following functions: $x = 1$, $y = f(x) = x - 1$, and $y = g(x) = -x + 5$. Sketch (i) the solid, (ii) the region, and (iii) a typical washer for this object.
NOTE: you only need to write the integral(s) but you do not need to compute it!

$0 \leq y \leq 2$ $r_o: y = x - 1 \Rightarrow x = y + 1$
 $r_i: x = 1$

$2 \leq y \leq 4$ $r_o: y = -x + 5 \Rightarrow x = -y + 5$
 $r_i: x = 1$

$$V = \int_0^2 \pi (r_o^2 - r_i^2) dy + \int_2^4 \pi (r_o^2 - r_i^2) dy$$



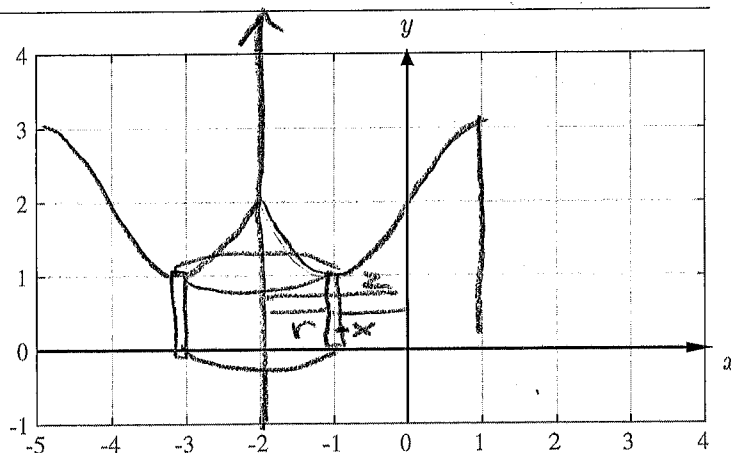
$$V_1 = \int_0^2 \pi [(y+1)^2 - 1] dy + \int_2^4 \pi [(-y+5)^2 - 1] dy$$

4. (10 pts) Using volumes by **SHELLS**, write an explicit integral for the solid generated by rotating about the line $x = -2$ (note that the line is off-axis!) the region delimited by the graphs of: $y = 2 + \sin(\frac{\pi}{2}x)$ and $y = 0$ for $-2 \leq x \leq 1$. (a) Sketch the functions, (b) the region, (c) the solid and (d) a typical shell for this object.
Note: you only need to write the integral but you do not need to compute it!

$$V = \int_{x_1}^{x_2} 2\pi r \cdot \text{height} dx$$

$$r - x = 2 \Rightarrow r = x + 2$$

$$= \int_{-2}^1 2\pi (x+2) [2 + \sin(\frac{\pi}{2}x)] dx$$



$$V_2 = \int_{-2}^1 2\pi (x+2) [2 + \sin(\frac{\pi}{2}x)] dx$$

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5. (8 pts) (a) Filling the boxes below, describe the volume of revolution that leads, using the method of **WASHERS**, to the integral:

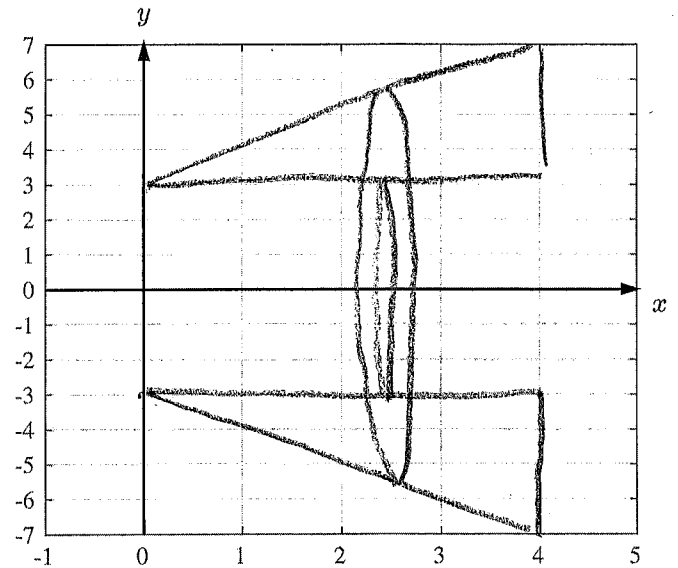
$$I = \int_0^4 \pi [(x+3)^2 - 9] dx.$$

- (b) Sketch the functions, the solid and
(c) a typical washer for this object.

The integral I above describes the volume generated by rotating about the $y = 0$ axis, the region

delimited by the functions $y = x+3$ and

$y = 3$ for $0 \leq x \leq 4$.



6. (10 pts) Show, using the method of **VOLUME BY SHELLS**, that the volume of a cone with circular base of radius R and height H is given by $V = \frac{1}{3}\pi R^2 H$. Draw a diagram including a typical slice for this object. Clearly indicate the function(s) that you are plotting and the interval of integration.

See v1

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7. (10 pts) Work.

- a) (3 pts) Compute the work done by the force $F(x) = kx$ (i.e., a linear, Hooke's, spring) when moving an object from $x = x_0$ to $x = x_1$.

See V1

- b) (3 pts) If one can only afford to use 24 Joules (Joule is the metric unit for energy: $1 \text{ J} = 1 \text{ kg m}^2/\text{s}^2$), how far is it possible to drag an object to the right, starting at $x = 0$ whose drag force is given by $F(x) = 2 + 2x$? Simplify as much as possible and leave your result as a single number with the correct units.

$$\begin{aligned} F &= 24 \Rightarrow \int_0^D (2+2x) dx = 24 \Rightarrow [2x+x^2]_0^D = 24 \\ \Rightarrow 2D+D^2 &= 24 \Rightarrow D^2+2D-24=0 \\ \Rightarrow D &= \frac{-2 \pm \sqrt{4+4 \cdot 24}}{2} = \frac{-2 \pm \sqrt{4+96}}{2} = \frac{-2 \pm \sqrt{100}}{2} \\ \therefore D &= \frac{-2 \pm 10}{2} = \frac{-2-10}{2}, \frac{-2+10}{2} = (-6, 4) \\ \text{since } D > 0 &\Rightarrow \boxed{D = 4 \text{ m}} \end{aligned}$$

- c) (4 pts) A sack with 50 Kg of sand is being lifted from $y = 0$ m to $y = 30$ m. The sack has a small hole that allows 100 g of sand to be lost every meter that the sack is pulled up. Compute the total energy necessary to lift the sack.
Hint#1: $F = m \times g$, use $g \approx 10 \text{ m/s}^2$, $1 \text{ Kg} = 1,000 \text{ g}$.
Hint#2: First compute the mass as a function of height.

$$\begin{aligned} m(y) &= 50 - 0.1y \\ W &= \int_0^{30} F = \int_0^{30} mg = \int_0^{30} 10 [50 - 0.1y] dy \\ &= \int_0^{30} (500 - y) dy = [500y - \frac{y^2}{2}]_0^{30} = 15000 - \frac{900}{2} \\ &= 15000 - 450 \\ \Rightarrow \boxed{W = 14550 \text{ J}} \end{aligned}$$

Do NOT write ANYTHING above this line!

8. (20 pts) Compute the following integrals

a) (4 pts) $I_1 = \int \cos^2(x) \sin^5(x) dx = \int \cos^2 \underbrace{\sin^4}_{du \Rightarrow u = -\cos} dx = \int \cos^2 (\sin^2)^2 dx$

$$= \int \cos^2 (1 - \cos^2)^2 dx = \int (u)^2 (1 - (u)^2)^2 du = \int u^2 (1 - u^2)^2 du$$

$$= \int u^2 (1 - 2u^2 + u^4) du = \int (u^2 - 2u^4 + u^6) du = \frac{1}{3}u^3 - \frac{2}{5}u^5 + \frac{1}{7}u^7 + C$$

or $-\frac{1}{3}\cos^3 + \frac{2}{5}\cos^5 - \frac{1}{7}\cos^7$

$$I_1 = \frac{1}{3}(-\cos x)^3 - \frac{2}{5}(-\cos x)^5 + \frac{1}{7}(\cos x)^7 + C$$

Not necessary

b) (5 pts) $I_2 = \int 2x^2 \cos(x) dx$

Same as V2 but $\div 2$

$$I_2 = -2x^2 \sin x + 4x \cos x + 4 \sin x + C$$

Not necessary

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c) (3 pts) $I_3 = \int 5x \sinh(3x^2 + 3) dx = \int \sinh(u) \frac{9}{5} du = 9/5 \cosh u + C$

U-sub : $u = 3x^2 + 3$
 $du = 6x dx \Rightarrow 5x dx = \frac{5}{6} du$
~~Part~~

$I_3 = 9/5 \cosh(3x^2 + 3) + C$

Not necessary

d) (5 pts) $I_4 = \int_0^2 x e^{x+3} dx = [x e^{x+3}]_0^2 - \int_0^2 e^{x+3} dx$

Part : $u = x \Rightarrow du = dx$
 $v = e^{x+3} \Rightarrow dv = e^{x+3} dx$

$= [x e^{x+3}]_0^2 - [e^{x+3}]_0^2 = (2e^5 - 0e^3) - (e^3 - e^3)$
 $= 2e^5 - e^3 + e^3$

$I_4 = e^5 + e^3$

e) (3 pts) $I_5 = \int \sin^4(t) dt$

double angle : $\sin^4(t) = (\sin^2(t))^2 = \left(\frac{1 - \cos 2t}{2}\right)^2$
 $= \frac{1}{4} [1 - 2 \cos 2t + \cos^2 2t]$

$\Rightarrow I_5 = \frac{1}{4} \int [1 - 2 \cos 2t + \cos^2 2t] dt = \frac{1}{4} \left[t - \sin 2t + \int \frac{1 + \cos 4t}{2} dt \right]$

but $\cos^2 2t = \frac{1 + \cos 4t}{2}$

$\Rightarrow I_5 = \frac{1}{4} \left[t - \sin 2t + \frac{1}{2} t + \frac{1}{2 \times 4} \sin 4t \right]$
 $= \frac{3}{8} t - \frac{1}{4} \sin 2t + \frac{1}{32} \sin 4t$

$I_5 = 3/8 t - 1/4 \sin 2t + 1/32 \sin 4t + C$

Not necessary

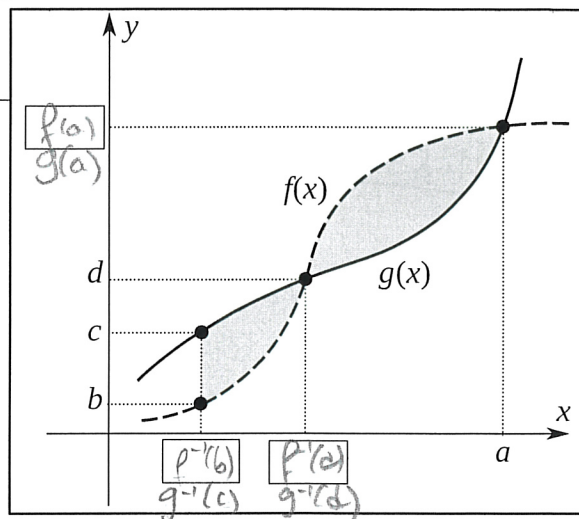
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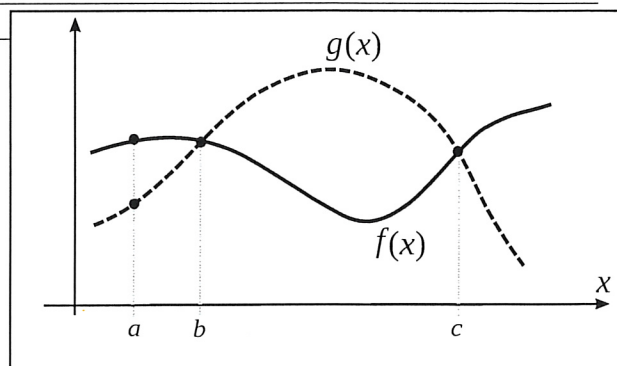
$$A_x = \int_{f^{-1}(b)}^{f^{-1}(d)} (g(x) - f(x)) dx + \int_{f^{-1}(d)}^a (f(x) - g(x)) dx$$

$$A_y = \int_b^c (f^{-1}(y) - g^{-1}(b)) dy + \int_c^d (f^{-1}(y) - g^{-1}(y)) dy + \int_d^{g(a)} (g^{-1}(y) - f^{-1}(y)) dy$$

2. (4 pts) Applications of integrals: averages.

For the functions depicted to the right, write the corresponding integrals for the **average vertical distance** between the graphs of the functions. Perform this for the following intervals:

- (a) Average vertical distance on $a \leq x \leq b$
 (b) Average vertical distance on $b \leq x \leq c$
 (c) Average vertical distance on $a \leq x \leq c$



OR using 1.1.

(a) ave. on $[a, b]$: $\frac{1}{b-a} \int_a^b (f(x) - g(x)) dx$

(b) ave. on $[b, c]$: $\frac{1}{c-b} \int_b^c (g(x) - f(x)) dx$

(c) ave. on $[a, c]$: $\frac{1}{c-a} \left[\int_a^b (f(x) - g(x)) dx + \int_b^c (g(x) - f(x)) dx \right]$

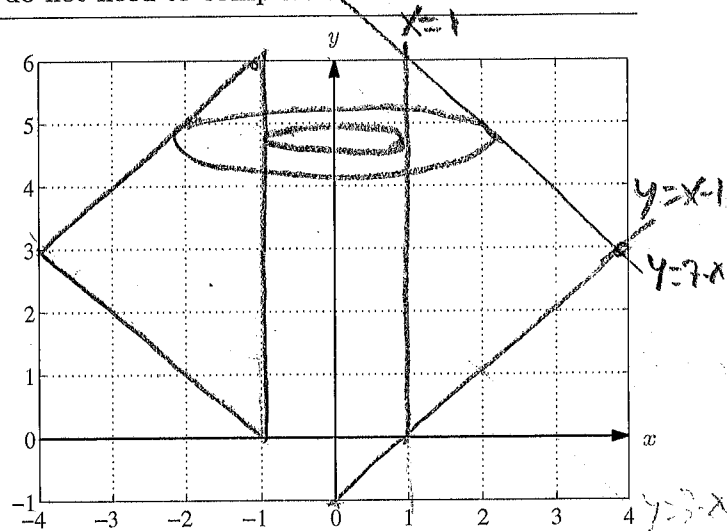
Do NOT write ANYTHING above this line!

3. (10 pts) Using **WASHERS** write an integral (or integrals) for volume of the solid generated by rotating about the y -axis the region inside the graphs of the following functions: $x = 1$, $y = f(x) = x - 1$, and $y = g(x) = -x + 7$. Sketch (i) the solid, (ii) the region, and (iii) a typical washer for this object.

NOTE: you only need to write the integral(s) but you do not need to compute it!

$0 \leq y \leq 3$ $f_0: y = x - 1 \Rightarrow x = y + 1$
 $R = 1$
 $3 \leq y \leq 6$ $f_0: y = 7 - x \Rightarrow x = 7 - y$
 $r = 1$

$$V = \int_0^3 \pi (R^2 - r^2) dy + \int_3^6 \pi (R^2 - r^2) dy$$



$$V_1 = \int_0^3 [(y+1)^2 - 1] dy + \int_3^6 [(7-y)^2 - 1] dy$$

4. (10 pts) Using volumes by **SHELLS**, write an explicit integral for the solid generated by rotating about the line $x = -2$ (note that the line is off-axis!) the region delimited by the graphs of: $y = 2 + \cos(\frac{\pi}{2}x)$ and $y = 0$ for $-2 \leq x \leq 1$. (a) Sketch the functions, (b) the region, (c) the solid and (d) a typical shell for this object.

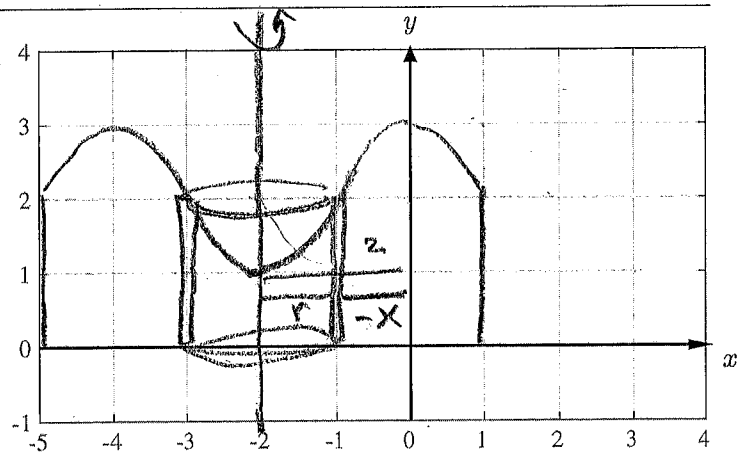
Note: you only need to write the integral but you do not need to compute it!

$$V = \int_{x_1}^{x_2} 2\pi r \cdot \text{height} \cdot dx$$

$$= \int_{-2}^1 2\pi r \cdot \text{height} \cdot dx$$

$$r - x = 2 \Rightarrow r = 2 + x$$

$$= \int_{-2}^1 2\pi (2+x) [2 + \cos(\frac{\pi}{2}x)] dx$$



$$V_2 = \int_{-2}^1 2\pi (2+x) [2 + \cos(\frac{\pi}{2}x)] dx$$

Do NOT write ANYTHING above this line!

5. (8 pts) (a) Filling the boxes below, describe the volume of revolution that leads, using the method of **WASHERS**, to the integral:

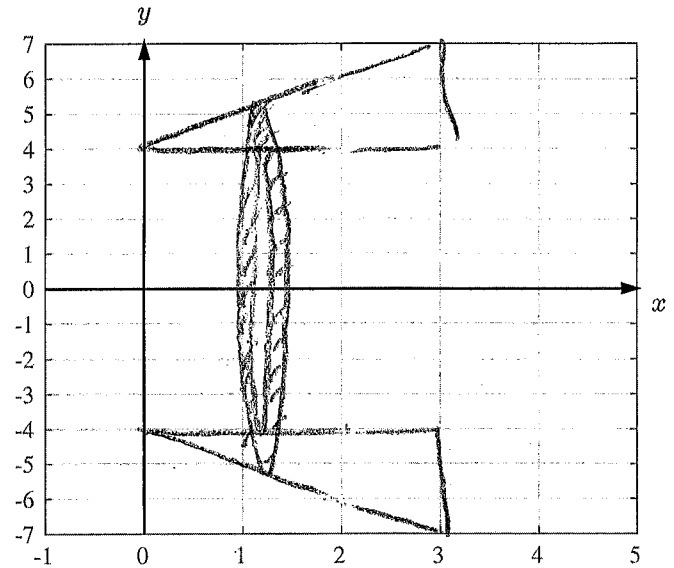
$$I = \int_0^3 \pi [(x+4)^2 - 16] dx.$$

- (b) Sketch the functions, the solid and
(c) a typical washer for this object.

The integral I above describes the volume generated by rotating about the $y = 0$ axis, the region

delimited by the functions $y = x + 4$ and

$y = 4$ for $0 \leq x \leq 3$.



6. (10 pts) Show, using the method of **VOLUME BY SHELLS**, that the volume of a cone with circular base of radius R and height H is given by $V = \frac{1}{3}\pi R^2 H$. Draw a diagram including a typical slice for this object. Clearly indicate the function(s) that you are plotting and the interval of integration.

See V1

Do NOT write ANYTHING above this line!

7. (10 pts) Work.

- a) (3 pts) Compute the work done by the force $F(x) = kx$ (i.e., a linear, Hooke's, spring) when moving an object from $x = x_0$ to $x = x_1$.

see V1

- b) (3 pts) If one can only afford to use 18 Joules (Joule is the metric unit for energy: $1 \text{ J} = 1 \text{ kg m}^2/\text{s}^2$), how far is it possible to drag an object to the right, starting at $x = 0$ whose drag force is given by $F(x) = 3 + 2x$? Simplify as much as possible and leave your result as a single number with the correct units.

$$F=18 \Rightarrow \int_0^D (3+2x) dx = 18 \Rightarrow [3x+x^2]_0^D = 18$$

$$\Rightarrow 3D+D^2=18 \Rightarrow D^2+3D-18=0$$

$$\Rightarrow D = \frac{-3 \pm \sqrt{9+4 \cdot 18}}{2} = \frac{-3 \pm \sqrt{81}}{2} = \frac{-3 \pm 9}{2}$$

$$\Rightarrow D = -\frac{3-9}{2}, -\frac{3+9}{2} = (-6, 3)$$

$$\text{Since } D > 0 \Rightarrow \boxed{D = 3 \text{ m}}$$

- c) (4 pts) A sack with 40 Kg of sand is being lifted from $y = 0 \text{ m}$ to $y = 5 \text{ m}$. The sack has a small hole that allows 100 g of sand to be lost every meter that the sack is pulled up. Compute the total energy necessary to lift the sack.

Hint#1: $F = m \times g$, use $g \approx 10 \text{ m/s}^2$, $1 \text{ Kg} = 1,000 \text{ g}$.

Hint#2: First compute the mass as a function of height.

$$m(y) = 40 - 0.1y$$

$$\begin{aligned} \Rightarrow \int_0^5 F(y) dy &= \int_0^5 mg dy = \int_0^5 (40 - 0.1y) \times 10 dy \\ &= \int_0^5 (400 - y) dy = \left[400y - \frac{y^2}{2} \right]_0^5 = 400 \times 5 - \frac{25}{2} \\ &= 2000 - 12.5 \end{aligned}$$

$$\Rightarrow \boxed{W = 1987.5 \text{ J}}$$

Do NOT write ANYTHING above this line!

8. (20 pts) Compute the following integrals

a) (4 pts) $I_1 = \int \cos^4(x) \sin^3(x) dx = \int \cos^4(x) \sin^2(x) \sin(x) dx = \int \cos^4(1-\cos^2) \sin(x) dx$
 $\quad \quad \quad du \Rightarrow u = -\cos$
 $= \int (u)^4 (1-(u)^2) du = \int u^4 (1-u^2) du = \int (u^4 - u^6) du$
 $= \frac{1}{5} u^5 - \frac{1}{7} u^7 + C$

OR $-\frac{1}{5} \cos^5 x + \frac{1}{7} \cos^7 x + C$

$I_1 = \frac{1}{5} (-\cos x)^5 - \frac{1}{7} (-\cos x)^7 + C$

Not necessary

b) (5 pts) $I_2 = \int 4x^2 \cos(x) dx = 4x^2 \sin x - \int 8x \sin x dx$

Part: $u = 4x^2 \rightarrow du = 8x dx$
 $v = \sin x \rightarrow dv = \cos x dx$

Part: $u = 8x \rightarrow du = 8 dx$
 $v = -\cos x \rightarrow dv = \sin x dx$

$= -4x^2 \sin x - [8x(-\cos x) - \int -\cos x \cdot 8 dx]$

$= -4x^2 \sin x + 8x \cos x + 8 \sin x + C$

Not necessary

$I_2 = -4x^2 \sin x + 8x \cos x + 8 \sin x + C$

Do NOT write ANYTHING above this line!

c) (3 pts) $I_3 = \int 3x \cosh(5x^2 + 2), dx = \int \cosh u \frac{3}{10} du = \frac{3}{10} \sinh u + C$

U-sub $u = 5x^2 + 2$
 $du = 10x dx \Rightarrow 3x dx = \frac{3 du}{10}$

~~Part 5~~

Not necessary

$$I_3 = \frac{3}{10} \sinh(5x^2 + 2) + C$$

d) (5 pts) $I_4 = \int_0^3 x e^{x+4} dx = [x e^{x+4}]_0^3 - \int_0^3 e^{x+4} dx = [x e^{x+4}]_0^3 - [e^{x+4}]_0^3$

Part: $u = x \rightarrow du = dx$
 $v = e^{x+4} \leftarrow dv = e^{x+4} dx$

$$\Rightarrow I_4 = 3e^7 - 0e^4 - [e^7 - e^4] \\ = 3e^7 - e^7 + e^4$$

$$I_4 = 2e^7 + e^4$$

e) (3 pts) $I_5 = \int \cos^4(t) dt$

see V1

$$I_5 =$$